

GOVERNMENT CITY COLLEGE

Hyderabad

Department of Computer Science

Course : BCA FINAL YEAR

Title : UniDesk Faculty Portal

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Abstract

The main objective of UniDesk is to develop a web-based faculty portal that brings important academic administration tasks together. It allows faculty members to take attendance using a student-name roster and maintain student records by course, group, and semester.

The project also provides tools for publishing circulars, managing campus events and holidays, and submitting faculty leave requests. HR users can view faculty records and approve or decline leave requests. Role-based access keeps faculty and HR responsibilities separate.

By bringing these workflows into one workspace, UniDesk aims to make routine faculty administration more organized and easier to manage. The application runs independently with locally stored data and is not connected to the college information system.

2. Problem statement

Faculty work often involves switching between separate lists and processes for attendance, student information, notices, calendar dates, and leave. UniDesk brings these routine tasks together behind a role-aware sign-in, with an emphasis on clear navigation and straightforward classroom roll call.

3. Aim and objectives

- Provide a single faculty-facing portal for everyday academic administration.
- Let teachers mark attendance from a name-first class roster organized by course, group, semester, subject, and date.
- Maintain student details and support publishing notices and calendar dates.
- Separate faculty self-service leave information from HR review and approval tasks.
- Provide a clean, responsive interface that is easy to understand during a project presentation.

4. Intended users and access

Faculty

Faculty sign in to view the overview, manage attendance and student records, publish circulars and events, and request leave. In My HR, a faculty member can see their own profile and leave requests; leave records for other teachers are not shown to them.

HR / Approver

The first HR account is registered during initial setup. HR can view the faculty directory and the leave approval queue, then approve or decline pending requests. The application restricts leave-status changes to the HR role.

5. Main modules

Overview

A dashboard with shortcuts and summary counts for students, circulars, upcoming events, and pending leave items.

Attendance

Select a course (BCA, B.Sc, or B.Com), group, semester (1-6), subject, and date. The roster lists student names with roll numbers as supporting details. Mark each student Present, Absent, or Late; entries are saved for the selected class and date.

Students

Add and manage student names and roll numbers, associated with a course, group, and semester. Search and filters help find records.

Circulars

Write and publish notices for all students or a selected course. Published items appear in the circular list.

Events & Holidays

Add dated campus events and holidays and browse dates by year. The interface includes Republic Day, Independence Day (15 August), and Gandhi Jayanti entries.

My HR / HR Management

Faculty see only their own details and leave requests. HR sees the teacher directory and can approve or decline leave requests.

6. Typical workflows

Taking attendance

- 1 Sign in with a faculty account.
- 2 Open Attendance and choose course, group, semester, subject, and date.
- 3 Review the class roster by student name and confirm the matching roll number where needed.
- 4 Set the status for each student to Present, Absent, or Late. The selected status is saved against the student, class, and date.
- 5 Return to the same class and date to review or update the recorded entries.

Submitting and reviewing leave

A faculty member opens My HR, enters the requested dates and an optional reason, and submits the request. Its status is initially Pending. An HR user opens HR Management and uses the approval controls on the request to approve or decline it. Faculty accounts can see the outcome for their own requests.

7. Application design and technology

The project is implemented as a lightweight web application with separate presentation, interaction, and server files:

- index.html provides the page structure and navigation.
- styles.css contains responsive visual styling and layout.
- app.js handles browser interactions and calls the server API.
- server.mjs implements the HTTP server, API routes, authentication, access checks, and data persistence.
- data/campus-desk.json stores application records in JSON format.

The interface is served locally at `http://localhost:4173`. The server uses Node.js ES modules and built-in HTTP and crypto libraries; no separate database server is required for the local setup.

8. Authentication, roles, and data handling

Account registration and sign-in are handled by the application server. Passwords are stored as salted script hashes rather than plain text. A successful sign-in creates an `HttpOnly, SameSite=Lax` session cookie. API routes require an authenticated session, and role checks protect HR approval actions. Sessions are held in server memory and end when the server restarts. Application records are stored locally in the project JSON file.

9. Running the application

- 1 Install Node.js version 20 or newer.
- 2 Open a terminal in the project folder.
- 3 Run: `npm start`.
- 4 Open `http://localhost:4173` in a browser.

- 5 Create a faculty account. During initial setup, select HR / Approver once to create the HR account; later HR access must be provided by the HR account holder.

10. Scope and limitations

- This is a standalone project and is not connected to a college information system or official attendance database.
- The included setup stores records in a local JSON file. For multi-device use, it would need secure hosting and a shared database.
- There is no institutional single sign-on, student self-service attendance marking, photo or face recognition, QR attendance, or biometric verification.
- Attendance history is stored by date and subject; institution-specific reporting, exports, and audit controls can be added in a future version.
- Session records are held in memory and reset when the server restarts. Production deployment would need persistent sessions, HTTPS, backups, and an operational security review.
- The calendar includes a few preset national holidays; institutions should maintain their own authoritative holiday calendar.

11. Possible future improvements

- Move records to a managed database and add backups and data export.
- Add timetable integration and class reminders.
- Add attendance reports by student, class, subject, and date range.
- Add controlled account provisioning, password reset, and audit logs.
- Support institution-managed holiday calendars and richer circular targeting.
- Deploy on a secured HTTPS server so authorized staff can use a shared dataset.

12. Conclusion

UniDesk demonstrates how attendance, student records, circulars, campus dates, and faculty leave can be organized in one role-aware faculty portal. Its name-first roster and separate faculty and HR views support a clear attendance and leave workflow. The current version is a locally runnable project prototype; connection to institutional systems and production-scale shared data would require additional deployment and integration work.